

# Project Workflow

## 1. Data Collection

- Collected sales data and stored it in an Excel file (`sales_dataset.xlsx`).
- Dataset contains:
  - Product Name
  - Region
  - Sales Amount
  - Profit
  - Order Details

### Input

```
sales_dataset.xlsx
```

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## 2. Data Loading

- Used the Pandas library to load the Excel dataset into a DataFrame.
- Converted raw Excel data into a structured format for analysis.

### Code

```
df = pd.read_excel("sales_dataset.xlsx")
```

### Output

```
DataFrame
```

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## 3. Data Processing

- Processed the dataset using Pandas.
- Calculated key business metrics.

### KPIs Generated

#### Total Sales

```
total_sales = df["Sales"].sum()
```

#### Total Profit

```
total_profit = df["Profit"].sum()
```

#### Sales by Region

```
sales_region = df.groupby("Region")["Sales"].sum()
```

### Product-wise Sales

```
product_sales = df.groupby("Product")["Sales"].sum()
```

### Purpose

Transform raw business data into meaningful insights.

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## 4. Dashboard Development

- Developed an interactive dashboard using Streamlit.
- Displayed the complete dataset.
- Created KPI cards for sales and profit.

### Dashboard Components

#### Dataset Table

```
st.dataframe(df)
```

#### KPI Metrics

```
st.metric()
```

#### Output

```
Total Sales  
Total Profit
```

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## 5. Data Visualization

- Visualized sales performance by region.
- Created interactive charts to help users identify business trends.

### Chart

```
st.bar_chart(sales_region)
```

### Benefits

- Quick understanding of regional performance.
  - Easy comparison across regions.
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## 6. Natural Language Query System

- Added a question-answer interface.
- Users can ask business-related questions using natural language.

### Examples

Which region has highest sales?

What is total sales?

What is total profit?

Which is the best product?

### Input Method

```
st.text_input()
```

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## 7. Query Processing

- Converted user questions into lowercase text.
- Matched keywords with predefined business logic.

### Example

```
if "highest sales" in q:
```

### Purpose

Identify user intent and determine the required analysis.

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## 8. Business Insight Generation

The system performs analytical operations on the dataset.

### Example

Finding Best Region

```
region = sales_region.idxmax()
```

### Example Output

South Region has highest sales.

## Business Value

Provides instant answers without manually exploring dashboards.

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## 9. LLM Integration

- Integrated Phi-3 Mini through Ollama.
- Used an LLM to generate business recommendations.

### Flow

```
Dataset
↓
Prompt Engineering
↓
Phi-3 Mini
↓
Recommendations
```

### Prompt Example

```
prompt = f"""
You are a business analyst.
Give 3 short business recommendations.
"""
```

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## 10. AI Recommendation Engine

- Sends processed dataset information to Phi-3 Mini.
- LLM analyzes patterns and generates recommendations.

### Example Output

1. Focus on low-performing regions.
2. Increase stock for best-selling products.
3. Improve marketing efforts to boost profits.

### Business Benefit

Provides decision-making support to managers.

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## 11. Executive Summary Generation

- Added a one-click summary generation feature.

- Automatically prepares a management-level report.

## **Includes**

- Total Sales
- Total Profit
- Best Region
- Weakest Region
- Best Product
- Recommendations

## **Output Example**

Executive Summary

Total Sales: ₹5,00,000

Total Profit: ₹1,20,000

Best Region: South

Weakest Region: East

Best Product: Laptop

Recommendations:

- Improve East region sales.
  - Promote best-selling products.
  - Increase profit margins.
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# **12. User Interface Layer**

Built using Streamlit.

## **Features**

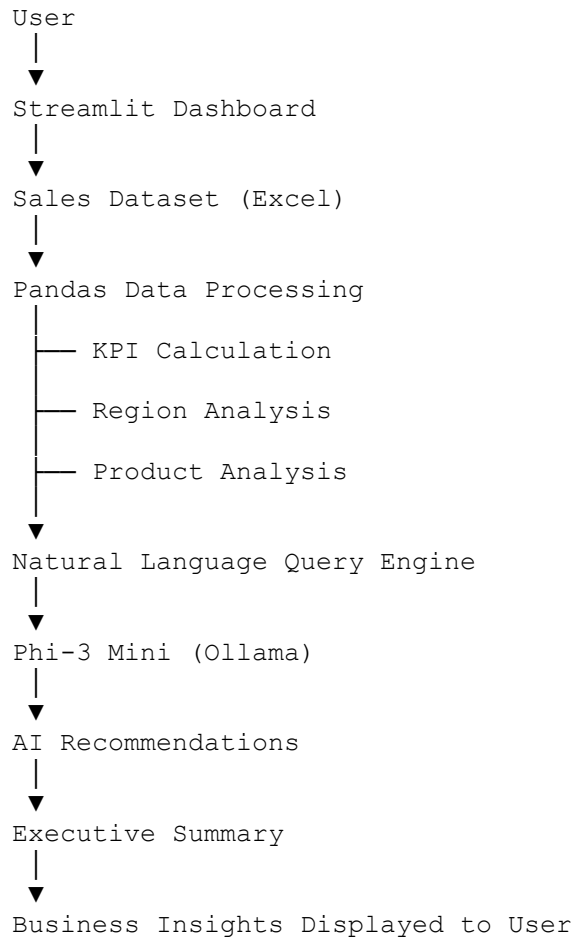
- Interactive dashboard
- KPI cards
- Business charts
- AI chatbot
- Summary generation

## **User Experience**

Simple and intuitive interface for non-technical users.

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# Complete Architecture



## Technologies Used

- Python
- Streamlit
- Pandas
- Ollama
- Phi-3 Mini (LLM)
- Excel

## One-Line Project Description :

**Developed an AI-Powered Business Intelligence Assistant that combines data analytics, interactive dashboards, natural language querying, and LLM-driven business recommendations using Streamlit, Pandas, Ollama, and Phi-3 Mini.**